

Material Classes

14 May 2025

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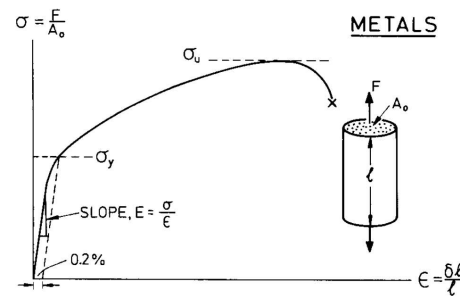
Metals and Alloys

Characterised by:

- Relatively **high** moduli
- **Ductile**, so can be formed
- Even high strength alloys, with ductility as low as **2%** yields enough to ensure failure is a **tough ductile type**.
- Partly because of ductility, prone to **fatigue**.
- Some have poor corrosion resistance.

Strengthened by:

- Alloying
- Deformation
- Heat treatment



Ceramics & Glasses

Characteristics:

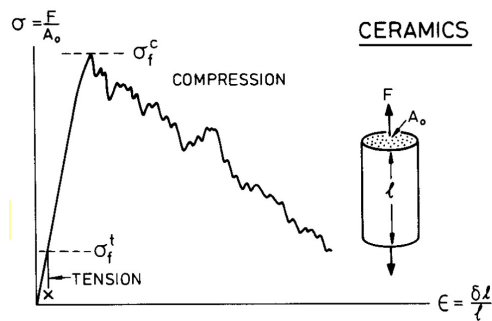
- Typically **high** moduli
- **Brittle** (No ductility)
- Strength in **tension** = Brittle fracture strength
- Strength in **compression** = Brittle crushing strength (**$\sim 10\times$**)
- Low tolerance for stress concentrations (**holes or cracks**) or high contact stresses (**at clamping points**)
- Scatter in strength depends upon volume of material under load
- **Harder to design with** than metals
- **Abrasion resistant** (Bearings and cutting tools)
- **High temperature capability**
- **Corrosion resistant**

Polymers

Characteristics

- Low moduli (**$\sim 1000\times$ less** than metals)

$E \downarrow +$



Composites

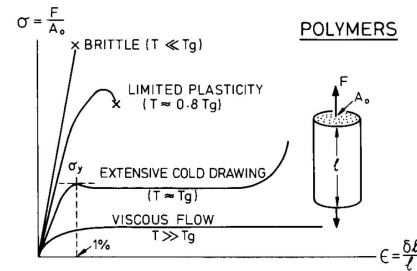
- Combination of 2 materials to exploit most attractive properties
- Typically polymer matrix (epoxy or polyester), usually reinforced by fibres such as glass, carbon fibre or Kevlar.

Characteristics:

- Light
- Stiff

Characteristics

- Low moduli (~1000x less than metals)
- ~5x lighter than metals.
- Some nearly as strong as metals
- Large elastic deflections
- Creep easily, even at room temp, so over time may reach permanent set if under load.
- Properties depend on temperature
- NONE have useful strength above 200°C
- Easy to shape / form
- Large elastic deflections allow for snap fixtures
 - Fast & Cheap assembly
- No finishing operations needed if mould is accurate.
- Corrosion resistance
- Low coefficients of friction.

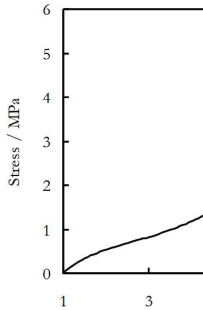


Elastomers

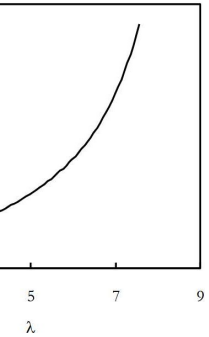
- Formed by crosslinking polymers.
- Have many properties of polymers, including:
 - high extensibility
 - Low modulus ($\sim \frac{1}{100000}$ th of metal)

4 requirements

- Temperature above glass transition temperature (T_g)
- Forces between molecules must be weak like in liquid.
- Polymer must not be highly crystalline
- Polymer must be lightly crosslinked



→ Velocity of elastic waves in a material proportional to $\left(\frac{E}{\rho}\right)^{1/2}$



- Light
- Stiff
- Strong
- Tough
- Polymer composites cannot be used above 250°C
- Expensive
- Difficult to form and join
- Used if added performance justifies added cost